OOP PROJECT REPORT

VEHICLE RENTLE SYSTEM

1. Introduction

The Vehicle Rental System is a console-based application developed using C++ programming language. It follows object-oriented programming principles to simulate a real-world vehicle rental scenario. This system allows customers to view available vehicles, rent them for a specific number of days, and generate an invoice automatically.

2. Problem Statement

Traditional vehicle rental systems rely on manual record keeping which can lead to errors, delays, and mismanagement. Calculating rent manually is time-consuming and error-prone. There is a need for a computerized solution to overcome these limitations.

3. Objectives of the System

- To implement core OOP concepts in C++
- To automate vehicle rental process
- To reduce human errors
- To provide accurate billing
- To create reusable and maintainable code

4. System Overview

The system consists of multiple classes that represent real-world entities such as Vehicle, Car, Bike, Customer, RentalCompany, Invoice, and Rental. Each class is designed to perform a specific role in the system.

5. OOP Concepts Implementation

Inheritance: Car and Bike classes inherit from Vehicle class.

Polymorphism: Virtual functions allow dynamic binding.

Association: Customer interacts with Vehicle objects.

Aggregation: RentalCompany contains vehicle objects.

Composition: Rental class owns Invoice object.

6. Tools and Technologies Used

- Programming Language: C++
- IDE: Code::Blocks / Dev-C++
- Paradigm: Object-Oriented Programming

7. Advantages of the System

- Simple and user-friendly
- Accurate rent calculation
- Easy to extend
- Real-world application of OOP

8. Limitations

- Console-based interface
- No permanent data storage
- Limited number of vehicles

9. Future Enhancements

- Add file handling for data storage
- Implement GUI interface
- Add database connectivity

- Add more vehicle types

10. Conclusion

The Vehicle Rental System successfully demonstrates the practical use of object-oriented programming concepts in C++. It provides a strong foundation for understanding real-world software design and can be enhanced further for commercial use.